MetaboAnnotation: simplifying metabolite annotation

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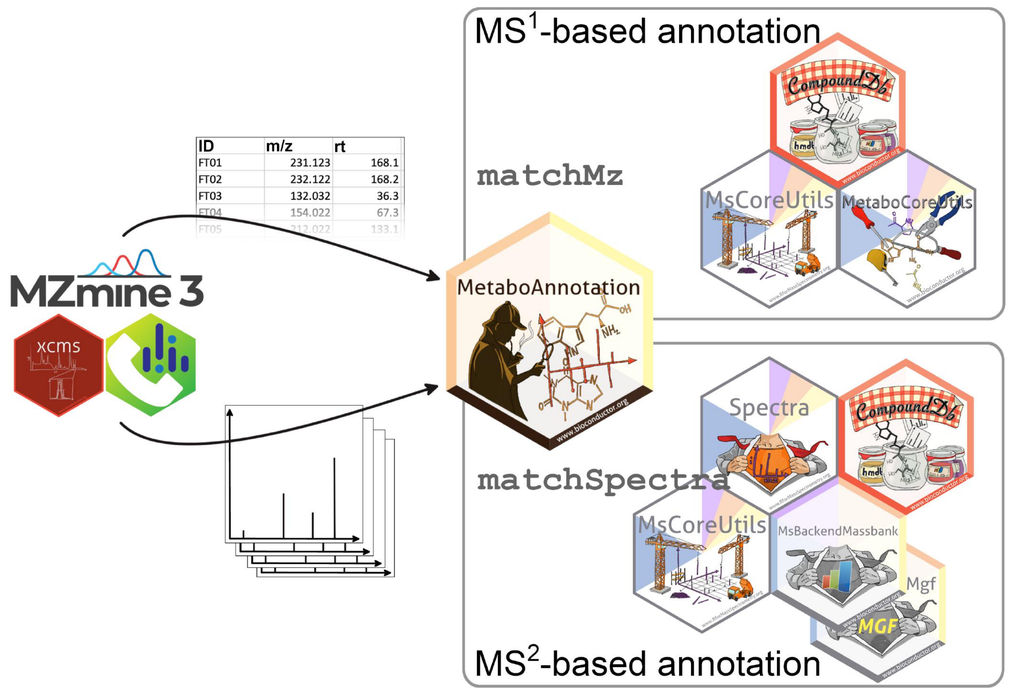
Introduction

`MetaboAnnotation`: Main end-user interface of a comprehensive ecosystem of R packages for MS[1]- and MS[2]-based metabolite annotation.



MS[1] annotation with matchValues()

`matchValues`: *m/z* and/or retention time-based annotation of LC-MS features.

Arguments:

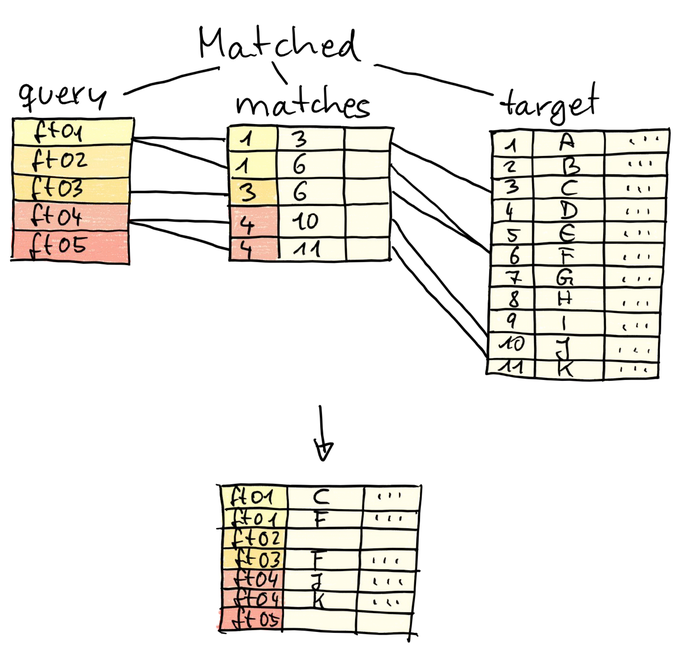
- `query`: Contains measured values.
- `target`: Contains reference values.
- `param`: Parameter object to select and configure the *matching* algorithm:
  - `MzParam` [`MzRtParam`]: Matching based on *m/z* values [and retention times]
  - `Mass2MzParam` [`Mass2MzRtParam`]: Matching based on *m/z* values [and retention times] after conversion of `target` masses to m/z values.
  - `Mz2MassParam` [`Mz2MassRtParam`]: Matching based on mass values [and retention times] after conversion of `query` and `target` *m/z* to mass values.

The variables `query` and `target` can be of type `numeric`, `data.frame`, `SummarizedExperiment` or `QFeatures`. These can be easily extracted from any `xcms` result objects or retrieved from `CompDb` or `IonDb` annotation databases.

Example: Adducts (for the mass to m/z conversion), absolute and m/z dependent tolerance for the matches are set via `param` object.

```
param <- Mass2MzParam(adducts = c("[M+H]+", "[M+Na]+"), tolerance = 0.005, ppm = 0)
matches <- matchValues(query, target, param)
```

Output: A result object of type `Matched` simplifies the handling (e.g. filtering, sub-setting, re-ordering and extracting matches) of matching results.



MS[2] annotation with matchSpectra()

`matchSpectra`: Compare and match experimental MS[2] spectra against reference spectra.

Arguments:

- `query`: A `Spectra` object containing experimental spectra.
- `target`: A `Spectra` object containing reference spectra (from in-house or external library).
- `param`: A parameter object to configure the matching algorithm (pre-filtering, similarity scoring algorithm, similarity cut-off value).
  - `CompareSpectraParam`: Identifies spectra with a similarity score above a user-defined threshold.
  - `MatchForwardReverseParam`: Same as above but in addition it also calculates the reverse score.

Example:

```
param = CompareSpectraParam(requirePrecursor = TRUE, ppm = 10)
matches <- matchSpectra(query, target, param)
```

Output: A result object of type `MatchedSpectra` that has the same structure as `Matched` and provides the same functionalities to simplify results handling.

An interactive, `shiny`-based, validation of matching results is available with the `validateMatchedSpectra` function:

```
validated_matches <- validateMatchedSpectra(matches)
```

valid	precursorMz	target_precursorMz	rtime	target_rtime	target_compound_name	score
true	300.029467324392	300.0301	361.7		Azaconazole	0.787
false	300.029467324392	300.0301	361.7		Azaconazole	0.886
true	300.029467324392	300.0301	361.7		Azaconazole	0.723
true	300.029467324392	300.0301	361.7		Azaconazole	0.722

Conclusions

- The functions `matchValues()` and `matchSpectra()` respectively facilitate MS[1] - and MS[2]-based annotation in R.
- The object classes `Matched` and `MatchedSpectra` help to handle complicated (can be one-to-many / one-to-none) matching results.
- The function `validateMatchedSpectra()` enables visual inspection and manual validation of the spectra matching results.
- We provide an expandable annotation ecosystem allowing easy integration of additional annotation resources and formats.

Reference: Rainer, J.; Vicini, A.; Salzer, L.; Stanstrup, J.; Badia, J.M.; Neumann, S.; Stravs, M.A.; Verri Hernandes, V.; Gatto, L.; Gibb, S.; Witting, M. *A Modular and Expandable Ecosystem for Metabolomics Data Annotation in R.* Metabolites 2022, 12, 173. https://doi.org/10.3390/metabo12020173

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